1 Fig. 7 shows a curve and the coordinates of some points on it.

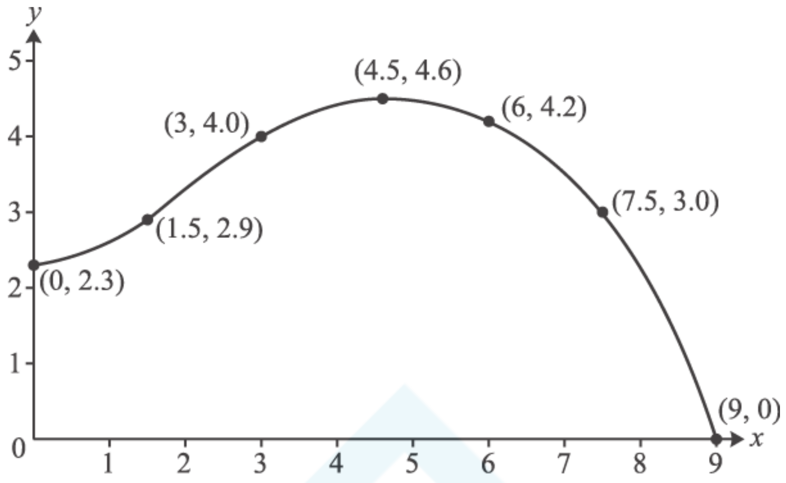


Fig. 7

Use the trapezium rule with 6 strips to estimate the area of the region bounded by the curve and the positive x- and y-axes.

[4]

2 Oskar is designing a building. Fig. 12 shows his design for the end wall and the curve of the roof. The units for x and y are metres.

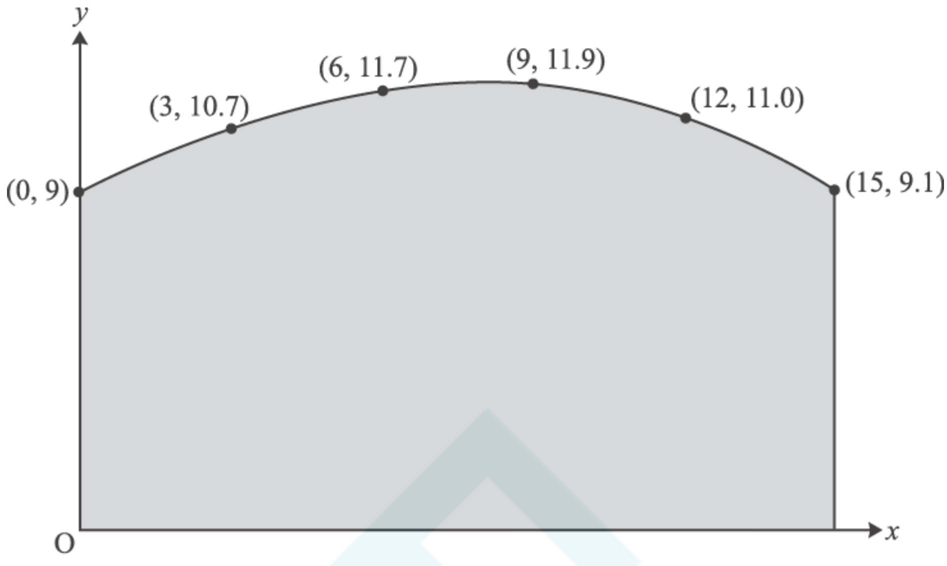


Fig. 12

(i) Use the trapezium rule with 5 strips to estimate the area of the end wall of the building.

[4]

(ii) Oskar now uses the equation $y = -0.001x^3 - 0.025x^2 + 0.6x + 9$, for $0 \le x \le 15$, to model the curve of the roof.

(*A*) Calculate the difference between the height of the roof when $x = 12$ given by this model and the data shown in Fig. 12.

[2]

(*B*) Use integration to find the area of the end wall given by this model.

[4]

3

Fig. 9.1

(i) Jean is designing a model aeroplane. Fig. 9.1 shows her first sketch of the wing's cross-section. Calculate angle A and the area of the cross-section.

[5]

(ii) Jean then modifies her design for the wing. Fig. 9.2 shows the new cross-section, with 1 unit for each of x and y representing one centimetre.

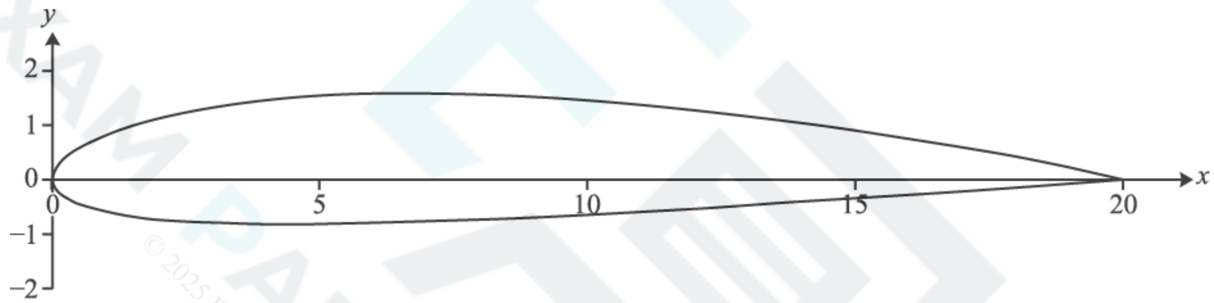


Fig. 9.2

Here are some of the coordinates that Jean used to draw the new cross-section.

Upper surface		Lower surface	
x	y	x	y
0	0	0	0
4	1.45	4	–0.85
8	1.56	8	–0.76
12	1.27	12	–0.55
16	1.04	16	–0.30
20	0	20	0

Use the trapezium rule with trapezia of width 4 cm to calculate an estimate of the area of this cross–section.

[6]

4 Fig. 9 shows the cross-section of a straight, horizontal tunnel. The x-axis from 0 to 6 represents the floor of the tunnel.

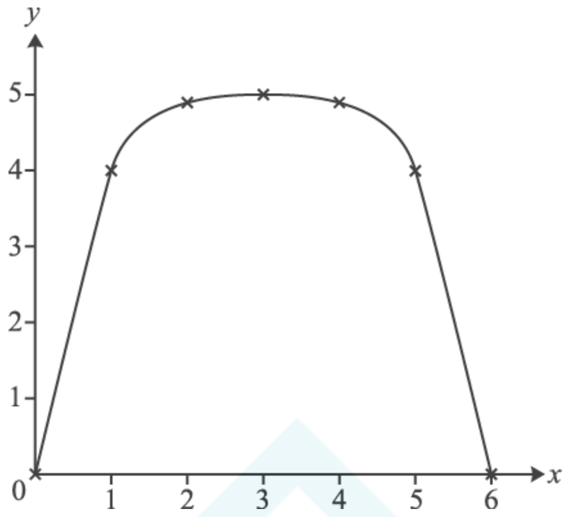


Fig. 9

With axes as shown, and units in metres, the roof of the tunnel passes through the points shown in the table.

x	0	1	2	3	4	5	6
y	0	4.0	4.9	5.0	4.9	4.0	0

The length of the tunnel is 50 m.

(i) Use the trapezium rule with 6 strips to estimate the area of cross-section of the tunnel. Hence estimate the volume of earth removed in digging the tunnel.

[4]

(ii) An engineer models the height of the roof of the tunnel using the curve . This curve is symmetrical about $x = 3$.

(A) Show that, according to this model, a vehicle of rectangular cross-section which is 3.6 m wide and 4.4 m high would not be able to pass through the tunnel.

[2]

(B) Use integration to calculate the area of the cross-section given by this model. Hence obtain another estimate of the volume of earth removed in digging the tunnel.

[5]

5 Fig. 3 shows the curve $y = x^3 + \sqrt{(\sin x)}$ for $0 \leqslant x \leqslant \frac{\pi}{4}$.

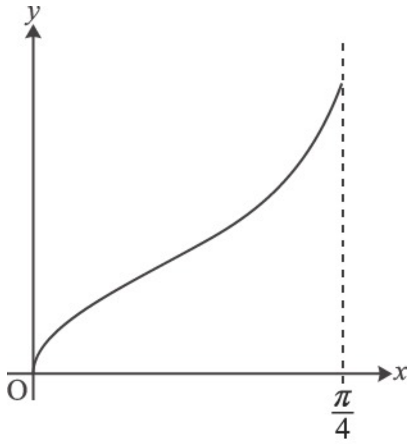


Fig. 3

(i) Use the trapezium rule with 4 strips to estimate the area of the region bounded by the curve, the *x*-axis and the line $x = \frac{\pi}{4}$, giving your answer to 3 decimal places.

[4]

(ii) Suppose the number of strips in the trapezium rule is increased. Without doing further calculations, state, with a reason, whether the area estimate increases, decreases, or it is not possible to say.

[1]

6 Fig. 6 shows a partially completed spreadsheet. This spreadsheet uses the trapezium rule with four strips to estimate

$$\int_0^{\frac{1}{2}\pi} \sqrt{1+\sin x}\, dx\ .$$

	A	B	C	D	E
1		x	$\sin x$	y	
2	0	0.0000	0.0000	1.0000	0.5000
3	0.125	0.3927	0.3827	1.1759	1.1759
4	0.25	0.7854	0.7071	1.3066	1.3066
5	0.375	1.1781	0.9239	1.3870	1.3870
6	0.5	1.5708	1.0000	1.4142	0.7071
7					5.0766
8					

Fig. 6

(a) Show how the value in cell B3 is calculated. [1]

(b) Show how the values in cells D2 to D6 are used to calculate the value in cell E7. [1]

(c) Complete the calculation to estimate $\int_0^{\frac{1}{2}\pi} \sqrt{1+\sin x}\,dx$, giving the answer to 3 significant figures. [2]

7 The function f(x) is defined by $f(x) = x^4 + x^3 - 2x^2 - 4x - 2$.

(a) Show that $x = -1$ is a root of $f(x) = 0$. **[1]**

(b) Show that another root of $f(x) = 0$ lies between $x = 1$ and $x = 2$. **[2]**

(c) Show that $f(x) = (x + 1)g(x)$, where $g(x) = x^3 + ax + b$ and a and b are integers to be determined. **[3]**

(d) Without further calculation, explain why $g(x) = 0$ has a root between $x = 1$ and $x = 2$. **[1]**

(e) Use the Newton-Raphson formula to show that an iteration formula for finding roots of g(x) = 0 may be written

$$x_{n+1} = \frac{2x_n^{\ 3} + 2}{3x_n^{\ 2} - 2}$$

Determine the root of g(x) = 0 which lies between x = 1 and x = 2 correct to 4 significant figures. **[3]**

8 Fig. 15 shows the graph of $f(x) = 2x + \frac{1}{x} + \ln x - 4$.

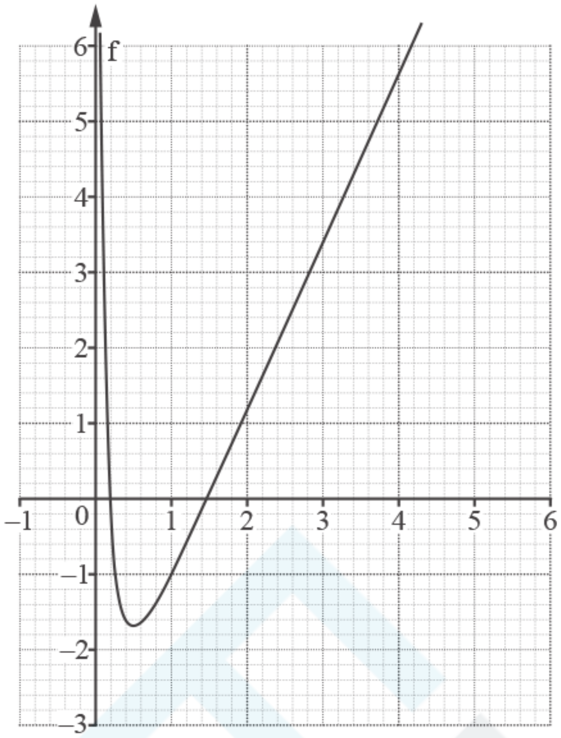


Fig. 15

(a) Show that the equation

$$2x + \frac{1}{x} + \ln x - 4 = 0$$

has a root, α, such that $0.1 < \alpha < 0.9$. **[2]**

(b) Obtain the following Newton-Raphson iteration for the equation in part **(a)**.

$$x_{r+1} = x_r - \frac{2x_r^3 + x_r + x_r^2(\ln x_r - 4)}{2x_r^2 - 1 + x_r}$$

[3]

(c) Explain why this iteration fails to find α using each of the following starting values.

(i) $x_0 = 0.4$ **[2]**

(ii) $x_0 = 0.5$ **[2]**

(iii) $x_0 = 0.6$ **[2]**

9 Fig. 3 shows the curve $y = \sqrt{1 + e^{2x}}$.

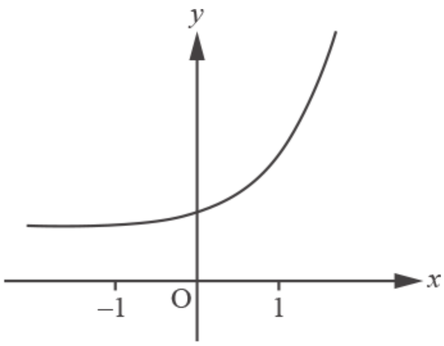


Fig. 3

The value of $\int_{-1}^{1} \sqrt{1 + e^{2x}}\,dx$ is to be estimated using the trapezium rule. T_2 and T_4 are the estimates obtained from the trapezium rule using 2 strips and 4 strips respectively.

(i) Explain whether T_4 is greater or less than T_2. [2]

(ii) Evaluate T_4, giving your answer to 3 significant figures. [4]

10 Joe uses the Newton-Raphson method to try to solve the equation $x^3 - 3x^2 - 10x + 25 = 0$.

(a) Show that the formula Joe should use is $x_{n+1} = x_n - \dfrac{x_n^3 - 3x_n^2 - 10x_n + 25}{3x_n^2 - 6x_n - 10}$. **[1]**

(b) Joe uses $x_0 = 4$ in this formula to find a root and obtains the following values:

$$x_1 = 3.928\ 571\ 4,$$

$$x_2 = 3.924\ 992\ 8.$$

Joe states that the root must be 3.92 to 2 decimal places and argues that this is because both x_1 and x_2 begin with 3.92.

(i) Comment on the validity of Joe's argument. **[1]**

(ii) Use a sign change argument to show that Joe's statement is correct. **[2]**

The graph of $y = x^3 - 3x^2 - 10x + 25$ in Fig. 11 shows that there is a root between 2 and 3.

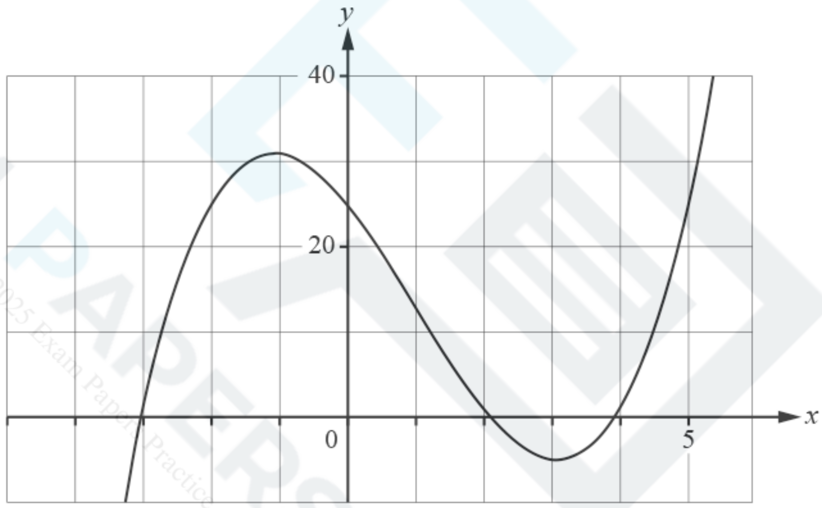


Fig.11

(c) Joe uses $x_0 = 3$ to attempt to find this root.

(i) Find x_1, x_2 and x_3. **[2]**

(ii) Explain why the Newton-Raphson method fails to give the required root in this case. **[2]**

(d) Explain what Joe should do to find the root of the equation between 2 and 3. **[1]**

11 By considering a change of sign, show that the equation $e^x - 5x^3 = 0$ has a root between 0 and 1. **[2]**

12

The function f (x) is defined by $f(x) = \sqrt[3]{27 - 8x^3}$. Jenny uses her scientific calculator to create a table of values for f (x) and f '(x).

x	f(x)	f'(x)
0	3	0
0.25	2.9954	–0.056
0.5	2.9625	–0.228
0.75	2.8694	–0.547
1	2.6684	–1.124
1.25	2.2490	–1.977
1.5	0	ERROR

(a) Use calculus to find an expression for f'(x) and hence explain why the calculator gives an error for f'(1.5). **[3]**

(b) Find the first three terms of the binomial expansion of f (x). **[3]**

(c) Jenny integrates the first three terms of the binomial expansion of f (x) to estimate the value of $\int_0^1 \sqrt[3]{27-8x^3}\,\mathrm{d}x$. Explain why Jenny's method is valid in this case. (You do not need to evaluate Jenny's approximation.)

[2]

(d) Use the trapezium rule with 4 strips to obtain an estimate for $\int_0^1 \sqrt[3]{27-8x^3}\,\mathrm{d}x$. **[3]**

The calculator gives 2.921 174 38 for $\int_0^1 \sqrt[3]{27 - 8x^3}\,dx$. The graph of y = f (*x*) is shown in Fig. 13.

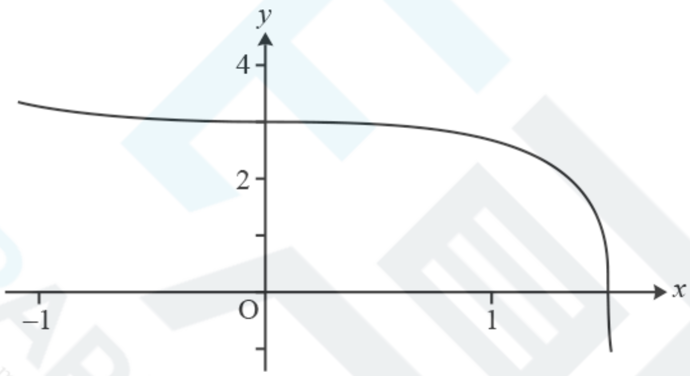


Fig. 13

(e) Explain why the trapezium rule gives an underestimate. [1]

Fig. 5.1 shows the curve $y = e^{1-x^2}$. Fig. 5.2 shows a spreadsheet used to calculate an estimate of $\int_0^2 e^{1-x^2}\,dx$ using the trapezium rule with four strips.

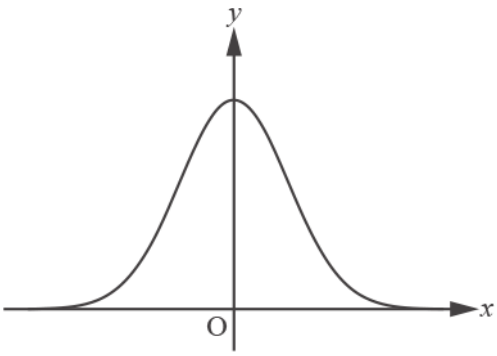


Fig. 5.1

	A	B	C
1	x		y
2	0	1	2.718282
3	0.5	0.75	2.117
4	1	0	1
5	1.5	-1.25	0.286505
6	2	-3	0.049787
7			

Fig. 5.2

(a) Show how the value in cell B3 is calculated. **[1]**

(b) Complete the calculation to estimate $\int_0^2 e^{1-x^2}\,dx$, giving the answer correct to 3 significant figures. **[2]**

(c) Show that the only stationary point on the curve is at (0, e). **[2]**

14 Rebecca is looking for the root of the equation

$$\sin 2x^2 - \cos 5x = 0$$

that lies between 0.2 and 0.3.

She uses the standard small angle approximations for $\sin \theta$ and $\cos \theta$ to find an estimate.

(a) Show that Rebecca's method gives an estimate of 0.26 when rounded to 2 decimal places. **[3]**

(b) Use a change of sign method to determine whether this value gives an estimate of the root which is correct to 2 decimal places. **[2]**

Rebecca then uses the Newton-Raphson method to find another estimate for the root.

(c) Show that using $x_0 = 0.2$ gives the value $x_1 = 0.291\,989\,3$ correct to 7 decimal places. **[5]**

(d) Continue this method, showing the result of each iteration, to find the root correct to 3 significant figures. **[3]**

Bob wishes to find an estimate for $\int_0^2 f(x)\,dx$, where $f(x) = \sqrt{x^{\frac{3}{2}} + 3}$, using the trapezium rule with 4 strips.

Fig. 6 is a screenshot of a spreadsheet Bob created to help him. In rows 2 to 6, the values in columns B and C have been multiplied to give the value in column D. The value in D7 is the sum of the values from D2 to D6.

	A	B	C	D
1	x	f(x)	multiplier	multiple of f(x)
2	0	1.732051	1	1.7321
3	0.5	1.831271	2	3.6625
4	1	2	2	4
5	1.5	2.199345	2	4.3987
6	2	2.414214	1	2.4142
7				16.2075

Fig. 6

(a) Calculate the estimate for $\int_0^2 \sqrt{x^{\frac{3}{2}} + 3}\,dx$ that Bob should obtain by using the trapezium rule with 4 strips. **[2]**

(b) You are given that the graph of $y = f(x)$ is concave upwards for $0 \le x \le 2$. Explain what you can deduce about the estimate for the integral obtained in part **(a)**. **[1]**

16(a) Fig. 14 shows a circle with centre O and radius r cm. The chord AB is such that angle AOB = x radians. The area of the shaded segment formed by AB is 5% of the area of the circle.

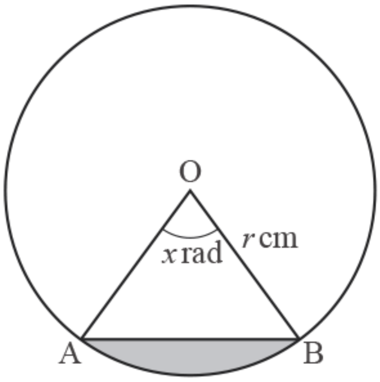


Fig. 14

[4]

Show that $x - \sin x - \frac{1}{10}\pi = 0$.

(b) The Newton-Raphson method is to be used to find x.

Write down the iterative formula to be used for the equation in part **(a)**. **[1]**

(c) Use three iterations of the Newton-Raphson method with $x_0 = 1.2$ to find the value of x to a suitable degree of accuracy. **[3]**

17 Fig. 11 shows the graph of $y = x^2 - 4x + x\ln x$.

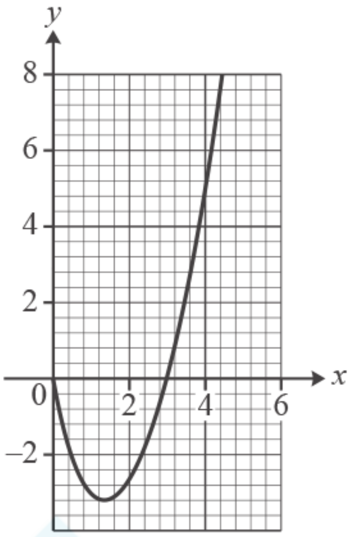


Fig. 11

Use an iterative method to find the x-coordinate of the stationary point on the curve $y = x^2 - 4x + x\ln x$, giving your answer correct to 4 decimal places. **[4]**

18

Find $\int \frac{1}{(x+2)(x+3)}\,dx$ in the form $\ln(f(x)) + c$, where c is the constant of integration and $f(x)$ is a function to be determined. **[3]**

19(a) **In this question you must show detailed reasoning.**

Express $\ln 3 \times \ln 9 \times \ln 27$ in terms of $\ln 3$. **[2]**

(b) Hence show that $\ln 3 \times \ln 9 \times \ln 27 > 6$. **[2]**

20(a) Fig. 8.1 shows the cross-section of a straight driveway 4 m wide made from tarmac.

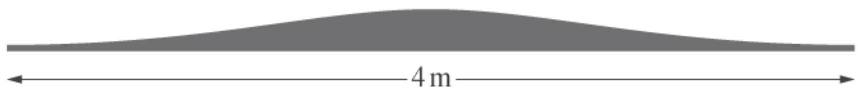


Fig. 8.1

The height h m of the cross-section at a displacement x m from the middle is modelled by $h = \frac{0.2}{1+x^2}$ for $-2 \leqslant x \leqslant 2$.

A lower bound of 0.3615 m^2 is found for the area of the cross-section using rectangles as shown in Fig. 8.2.

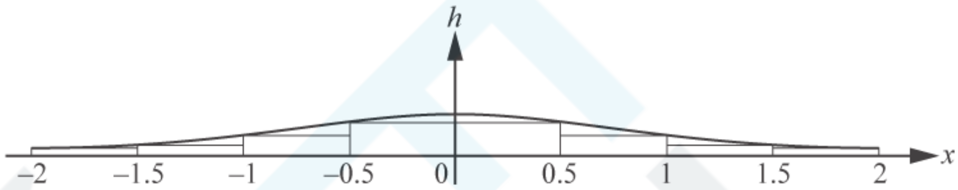


Fig. 8.2

Use a similar method to find an upper bound for the area of the cross-section. **[3]**

(b)

Use the trapezium rule with 4 strips to estimate $\int_0^2 \frac{0.2}{1+x^2}\,dx$. **[2]**

(c) The driveway is 10 m long. Use your answer in part **(b)** to find an estimate of the volume of tarmac needed to make the driveway. **[2]**

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Part marks and guidance	
1			h = 1.5	B1	h = 1.5	allow if used with 6 separate trapezia
			$\frac{1.5}{2}\times(2.3+2(2.9+4+4.6+4.2+3)+0)$	M1	basic shape of formula correct, omission of brackets may be recovered later	at least 4 y-values in middle bracket, e.g. $\frac{1.5}{2}\times(2.3+2(2.9+4+4.6+4.2)+3)$ **M0** if any x values used
			all y-values correct and correctly placed in formula	B1	condone omission of outer brackets and / or omission of 0	
			29.775 to 3 sf or better; isw	A1	answer only does not score <u>Examiner's Comments</u> A majority of candidates scored full marks, nearly always through correct application of the formula; although a few successfully used individual trapezia (the majority of those who adopted the latter approach were unsuccessful). Some slipped up by omitting the outer brackets and taking 3.0 (or occasionally 9) as the final y-value or by using an incorrect value for h (usually 1, occasionally 9 and rarely from incorrectly calculating 9÷6). Only a very small minority failed to score at all.	or **B1** + **B3** if 6 separate trapezia calculated to give correct answer
			Total	4		
2		i	h = 3 soi	B1		allow if used with 6 separate trapezia
		i	$\frac{3}{2}\{9+9.1+2(10.7+11.7+11.9+11.0)\}$	M1	basic shape of formula correct with their 3; omission of brackets may be recovered later; M0 if any x-values used (NB y_0 = 9 and x_3 = 9, so check position)	with 3, 4 or 5 y-values in middle bracket, eg $\frac{3}{2}\{9+2(10.7+11.7+11.9)+11.0\}$

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	all y-values correctly placed in formula	B1	condone omission of outer brackets	
		i	163.05 or 163.1 or 163 isw	A1	answer only does not score **Examiner's Comments** This was very well done, with many candidates scoring full marks. The most common error was the omission of the outer brackets; occasionally h = 5 was seen, and occasionally y values were misplaced.	or **B1** + **B3** if 5 separate trapezia calculated to give correct answer NB 29.55 + 33.6 + 35.4 + 34.35 + 30.15
		ii	(A) $-0.001 = 12^3 - 0.025 \times 12^2 + 0.6 \times 12 + 9$ soi	M1	may be implied by 10.872, 10.87 or 10.9	NB allow misread if minus sign omitted in first term if consistent in (A) and (B). Lose A1 in this part only
		ii	± 0.128[m] or ± 12.8cm or ± 128mm isw	A1	B2 if unsupported **Examiner's Comments** Many correctly substituted x = 12, and showed their working so that even if arithmetic went astray, a method mark was still earned. A common error was to omit the minus sign from the first term. Strangely many candidates stopped there, or subtracted 10.872 from 12 instead of 11.	appropriate units must be stated if answer not given in metres
		ii	$\mathrm{F}[x] = \frac{-0.001x^4}{4} - \frac{0.025x^3}{3} + \frac{0.6x^2}{2} + 9x$ (B)	M2	M1 if three terms correct; ignore + c	
		ii	F(15) [– F(0)] soi	M1	dependent on at least two terms correct in F[x]	condone F (15) + 0

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	161.7 to 162	A1	A0 if a numerical value is assigned to *c* **Examiner's Comments** This was generally very well done. Occasionally candidates made a sign error or inserted an extra zero in one or both of the first two terms. Some left “9” untouched or used an upper limit of 12 instead of 15.	answer only does not score NB allow misread if minus sign omitted in first term if consistent in (A) and (B). 187.03…
			Total	10		

Question			Answer/Indicative content	Marks	Part marks and guidance	
3		i	$[\cos A =]\dfrac{20^2+13^2-8^2}{2\times13\times20}$	M1*	or $8^2 = 20^2 + 13^2 - 2 \times 13 \times 20 \times \cos A$	
		i	$[\cos A =]\dfrac{505}{520}$ oe soi	A1	or 0.971 to 0.9712	
		i	A = 13.79 to 13.8° or 14°	A1	or 0.24077 to 0.241 or 0.24 (radians); allow **B3** if given to 3sf or more unsupported	or 15.32 (grad)
		i	[Area =] ½ × 20 × 13 × sin *their A*	M1dep*	or **M1** for eg ½ × 20 × 8 × sin 22.8, as long as angle calculated correctly from their A (other angles are 22.79824…° and 143.40645..° or 36.59355..°)	or $\sqrt{\frac{41}{2}(\frac{41}{2}-8)(\frac{41}{2}-13)(\frac{41}{2}-20)}$ NB 13sin A = 3.099899192 if ½ × b × h used

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	30.99 to 31.01 isw	A1	allow **B2** for unsupported answer within range **Examiner's Comments** Most candidates used the Cosine rule correctly to calculate angle A and most went on to calculate the area correctly using ½absinC. A minority complicated matters by splitting the triangle into 2 right angle triangles, calculating the height and then using Pythagoras to calculate the base of each triangle and hence the area of each separately. A few candidates unnecessarily found one of the other angles and then used ½absinC. In both cases this usually resulted in a loss of accuracy so their final answer was outside the permissible range. Similarly, some candidates rounded their value for cosA and went on to lose both accuracy marks. Approximately three quarters of candidates scored full marks.	
		i	or $\frac{5\sqrt{615}}{4}$ oe isw			
		ii	*alternatively*			
		ii	h = 4 soi	B1		
		ii	attempt to find all y-values	M1	$y_{upper} - y_{lower}$	**M0** if values are added to obtain 0.60, 0.80 etc
		ii	2.3, 2.32, 1.82, 1.34	A1	all y-values correct	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\frac{\text{their } 4}{2} \times (0 + 0 + 2(2.3 + 2.32 + 1.82 + 1.34))$	M1	shape of formula correct with 2, 3 or 4 of their *y* -values in inner bracket with their *h*; allow recovery from bracket errors **M0** if any non-zero *x*-values used or if *y*-values used twice	eg ½ × 4 × {2.3 + 1.34 + 2 (2.32 + 1.82)}
		ii		B1FT	all their *y*-values correctly placed, condone omission of zeros and/or omission of outer brackets	
		ii	31.12	A1	ignore subsequent rounding, but **A0** if answer spoiled by eg multiplication by 20 **Examiner's Comments** Most candidates used the trapezium rule separately for the upper and lower areas, with a smaller number recognising that the total area could be calculated with a single application. The most common error was omitting the outside pair of brackets in the formula and this was rarely recovered. Another common error was the failure to recognise that the absolute areas should be summed, with candidatessubtracting the lower area from the upper. A surprising error for several candidates was getting the value of h incorrect as this was not only clear from the table, but also explicitly stated in the question. Nevertheless, a significant majority scored full marks.	or **B1M1A1** + **B3** if area of 2 triangles and 3 trapezia calculated to give correct answer www NB 4.6 + 9.24 + 8.28 + 6.32 + 2.68

Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	11	

Question			Answer/Indicative content	Marks	Part marks and guidance	
4		i	$\frac{h}{2} \times (0 + 0 + 2[4 + 4.9 + 5 + 4.9 + 4])$ oe	M1	correct formula used with 4, 5 or 6 strips and numerical value for *h*; condone omission of zeros or omission of outer brackets for both **M** marks	allow eg ½ × 1 × (4 + 4 + 2[4.9 + 5 + 4.9]) ½ × 1 × (4 + 0 + 2[4 + 4.9 + 5 + 4.9]) (**NB** may be implied by 18.8 & 20.8 respectively)
		i	all non-zero *y*-values correctly placed	M1	**M0M0** if 1, 2, 3 or 6 used as *y*-values (these are *x* -values)	
		i	*h* = 1 used in formula or consistently with two triangles and four trapezia	B1	if **M0M0** allow **B1** for *h* = 1 and **B2** for 22.8 from area of 4 trapezia and 2 triangles and **B1** for 1140	
		i	area = 22.8 **and** volume = 1140 isw cao	A1	ignore units **Examiner's Comments** Most candidates used the Trapezium rule correctly and went on to score full marks. A few made bracket errors or misplaced the *y* -values. Even fewer successfully found the correct value for the area by splitting the area into separate triangles and rectangles. This approach is not recommended – most go wrong and fail to score.	if **M0M0B0** allow **SC4** for 22.8 and 1140 obtained correctly by other method
		ii	A substitution of *x* = 1.2 or 4.8 to find *y*	M1	allow substitution of $1.2 \le x \le 1.234$ or $4.766 \le x \le 4.8$	or **M1** for *y* = 4.4, *x* = 1.234 [or 4.766] and

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	y = 4.35 or 4.352 and correct comparison with 4.4 isw	A1	**Examiner's Comments** A minority worked out what to do here and used a correct value of x to find y, which was usually correctly compared with 4.4. However, many candidates misunderstood what was required, substitution of 3 or 3.6 were common errors. A few unsuccessfully tried to compare cross-sectional areas.	**A1** for comparison of 1.234 with 1.2 or 4.766 with 4.8 [so gap less than 3.6]
		ii	$\mathrm{F}[x]=\frac{5}{81}\left(\frac{108}{2}x^2-\frac{54}{3}x^3+\frac{12}{4}x^4-\frac{x^5}{5}\right)$ oe B	M2	**M1** for 3 correct terms; ignore $+c$	condone omission of $\frac{5}{81}$;
		ii	eg $\frac{10}{3}x^2-\frac{10}{9}x^3+\frac{5}{27}x^4-\frac{1}{81}x^5$		allow coefficients 3.333333…, 1.11111…, 0.185185…, 0.01234567…r.o.t to 2 sf or better or decimal equivalents in numerator: 6.6666…, 3.333333…, 0.74074…, 0.061728… r.o.t to 2 sf or better	**M0** if $\frac{5}{81}x$ seen outside braceket but next **M1** is still available; ignore subsequent attempt to evaluate c for first **M2**
		ii	F[6] – F[0] or 2 × (F[3] – F[0])	M1	dependent on at least two terms correctly integrated in bracket; condone omission of –F(0)	**M0** for non-zero lower limit
		ii	24	A1		24 unsupported does not score

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	1200	B1	**Examiner's Comments** Most candidates integrated successfully and substituted the correct limits to find the correct area. However, some made an error in one of the terms, and some omitted the factor of $^{5}/_{81}$, which cost the later accuracy marks A few candidates lost marks by substituting incorrect limits.	ignore units
			Total	11		

<table>
<tr><th colspan="3">Question</th><th>Answer/Indicative content</th><th>Marks</th><th colspan="2">Part marks and guidance</th></tr>
<tr><td>5</td><td></td><td>i</td><td><table><tr><td>x</td><td>0</td><td>0.1963</td><td>0.3927</td><td>0.5890</td><td>0.7854</td></tr><tr><td>y</td><td>0</td><td>0.4493</td><td>0.6792</td><td>0.9498</td><td>1.3254</td></tr></table></td><td>B2, 1,0</td><td>For values 0.4493,0.6792,0.9498 (4dp or better soi) [accept truncated to 4 figs after dec point]

[cannot assume values of form $(\pi/16)^3 + \sqrt{(\sin \pi/16)}$ are correct unless followed by correct total at some later stage as some will be in degree mode]</td><td></td></tr>
<tr><td></td><td></td><td>i</td><td>$A = (\pi/32)\ [(0 + 1.3254) + 2(0.4493 + 0.6792 + 0.9498)]$</td><td>M1</td><td>Use of the trapezium rule. Trapezium rule formula for 4 strips must be seen, with or without substitution seen. Correct h must be soi.</td><td></td></tr>
</table>

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	= 0.538	A1	[accept separate trapezia added] 0.538 **www 3dp only** (NB using 1.325 is ww) SC B0 0.538 without any working as no indication of strips or method used SC B1 0.538 with some indication of 4 strips but no values seen Correct values followed by 0.538 scores B2 B0 Correct values followed by correct formula for 4 strips, with or without substitution seen, then A = 0.538 scores 4/4. Correct formula for 4 strips and values of form $((\pi/16)^3$ + √(sinπ/16)… followed by correct answer scores 4/4 (or ¾ with wrong dp) **NB Values given in the table to only 3dp give apparently the correct answer, but scores B0,M1A0 ww** **Examiner's Comments** Many errors were seen here. In a number of cases the candidates were in degree mode. For others *h* was given incorrectly. Many others used the wrong formula and some substituted *x* values in the formula or omitted 0 from the formula. However, probably the most common error was giving the *y* values to 3dp and then using these to give a final answer correct to 3dp.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Not possible to say, eg some trapezia are above and some below curve oe.	B1	Need a reason. Must be without further calculation. **Examiner's Comments** This was a good discriminator as it really tested whether candidates understood how the trapezium rule estimates area. Some believed that it always underestimated or always overestimated.	
			Total	5		
6		a	A3*π oe	B1(AO2.2a) [1]	Or 0.125×π oe	
		b	$\frac{1}{2}$D2 + D3 + D4 + D5 + $\frac{1}{2}$ D6	B1(AO2.2a) [1]	Or equivalent expressed in words.	
		c	5.0766 × 0.3927 = 1.9935... 1.99 (units^2) (to 3sf)	M1(AO1.1) A1(AO1.1) [2]	Or 5.0766 × $\frac{\pi}{8}$	
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
7		a	$f(-1) = (-1)^4 + (-1)^3 - 2(-1)^2 - 4(-1) - 2$ $= 1 - 1 - 2 + 4 - 2 = 0$	E1(AO1.1) [1]		
		b	$f(1) = 1 + 1 - 2 - 4 - 2 = -6$ or 'negative' $f(2) = 16 + 8 - 8 - 8 - 2 = 6$ or 'positive' change of sign $\Rightarrow$ root between 1 and 2	B1(AO1.1) E1(AO2.4) [2]	both correct allow no mention of continuity of f AG	
		c	long division or equating coeffts $\Rightarrow g(x) = x^3 - 2x - 2$ so $a = -2$, $b = -2$	M1(AO1.1) A1A1(AO2.2a 1.1) [3]		
		d	Clear explanation E.g. $f(x) = (x + 1)g(x)$ For the root of $f(x) = 0$ between 1 and 2, RHS is also zero hence $g(x) = 0$	E1(AO2.4) [1]		
		e	$x_{n+1} = x_n - \frac{g(x_n)}{g'(x_n)}$ $= x_n - \frac{x_n^3 - 2x_n - 2}{3x_n^2 - 2}$ $= \frac{3x_n^3 - 2x_n - x_n^3 + 2x_n + 2}{3x_n^2 - 2}$ $= \frac{2x_n^3 + 2}{3x_n^2 - 2}$ Root 1.769 (4sf)	M1(AO1.1) E1(AO2.4) A1(AO2.2a) [3]	AG BC	
			Total	10		

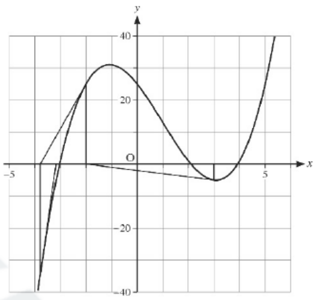
Question			Answer/Indicative content	Marks	Part marks and guidance	
8		a	f(0.1) = 3.897…and f(0.9) = –1.194… sign change so 0.1 < α < 0.9	B1(AO 1.1) E1(AO 1.2) [2]		
		b	$\left[\frac{dy}{dx}=\right] 2-\frac{1}{x^2}+\frac{1}{x}$ $x_{r+1}=x_r-\frac{2x_r+\frac{1}{x_r}+\ln x_r-4}{2-\frac{1}{x_r^2}+\frac{1}{x_r}}$ $x_{r+1}=x_r-\frac{2x_r^3+x_r+x_r^2(\ln x_r-4)}{2x_r^2-1+x_r}$	B1(AO 1.1) M1(AO 1.1) A1(AO 2.2a) [3]	AG	
		c	(A) x_1 = – 0.52359… which is negative so the iteration stops because f(x) is undefined for x < 0	B1(AO 1.1) E1(AO 2.4) [2]		
		c	(B) gradient is zero at 0.5 so tangent never touches x-axis	B1(AO 1.1) E1(AO 2.4) [2]		
		c	(C) 2.4497, 1.4667, 1.4675, 1.4675 *or* 0.6 is to the right of the turning point so converges to larger root	B1(AO 1.1) E1(AO 2.4) [2]		
			Total	11		

Question			Answer/Indicative content	Marks	Part marks and guidance	
9		i	$T_4 < T_2$ The approximation is an over-estimate, as the trapezia are above the curve therefore the error becomes less when the number of strips increases	B1 B1 [2]	oe (e.g. 4 T_4 is less than T_2) Must see mention of ‘over-estimate’ and ‘above’ and ‘increasing strips’ **Examiner’s Comments** The first mark in part (i) was awarded to the vast majority of candidates for correctly stating that T_4 was less than T_2 although some candidates did not make it explicitly clear which value of the two was the least. Candidates found the second mark a lot harder to come by as it was not sufficient to simply state that the approximations given by the trapezium rule were an over-estimate. Candidates needed to make it clear that these approximations were an over-estimate because the tops of the trapezia are above the curve which would then, in turn, mean that the error (in using a trapezium rule approximation for the value of the integral) would become less when the number of strips increases.	
		ii	0.5 × 0.5 {1.0655… + 2.8963… + 2(1.1695… + 1.4142… + 1.9282…)}	B1 M1 A1	For using 0.25 oe The M mark requires the correct {…} bracket structure. It needs the first bracket to contain the first y value plus the last y value and the second bracket to be multiplied by 2 and to be the summation of the	

Question			Answer/Indicative content	Marks	Part marks and guidance	
			$T_4 = 3.25$ OR 0.5587… + 0.6459… + 0.8356… + 1.2061…	A1 [4]	remaining y values with no additional values. Allow an error in one value **or** the omission of one value from the second bracket M0 if using x values. All values given to at least 3sf or exact The A mark is for the correct {…} bracket with no errors (12.98… or 13.0 implies M1A1) cao (3.25 with no working scores 0/4) – must be given to 2dp only (for reference correct answer is 3.2465079…) SC: bracketing error 0.25 × (1.0655 + 2.8963) + 2(1.1695… + …) scores B1M1A0A0 unless the final answer implies the correct calculation. An answer of 10.014… usually indicates this error Separate trapezia B1 – one area correct (implies 0.25) M1 – three correct (equivalent to one error) A1 – all four correct A1 – cao of 3.25 **Examiner's Comments** Part (ii) was answered extremely well with the vast majority of candidates giving the correct answer of 3.25. When errors occurred it was usually due to an incorrect value for the width of the strips or with the omission of a value. It was very rare for candidates to use the x values or to not give the answer to the required 3 significant figures.	

Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	6	

Question			Answer/Indicative content	Marks	Part marks and guidance	
10		a	$f(x) = x^3 - 3x^2 - 10x + 25 \Rightarrow$ $f'(x) = 3x^2 - 6x - 10$, so the N-R formula gives $x_{n+1} = x_n - \frac{x_n^3 - 3x_n^2 - 10x_n + 25}{3x_n^2 - 6x_n - 10}$	E1(AO2.1) [1]	AG Must be clear that denominator is derivative of numerator	
		b	(i) Not valid: the sequence may decrease further, far enough to change the first 3 figures	B1(AOs 2.3) [1]	Reason for 'not valid' needed	
			(ii) f (3.915) = – 0.1255… and f(3.925) = 2.03… x 10^{-4} Change of sign shows that there is a root in the interval (3.915, 3.925) so the root is 3.92 to 2dp	M1(AO2.1) A1(AO2.2a) [2]	Both calculations Complete argument needed	Allow use of any two values closer to 3.92 that give sign change

Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	(i $x_0 = 3$ &⇒ $x_1 = -2$) $x_2 = -3.785\ 71.$ and $x_3 = -3.168\ 34.$ (ii)  The initial value is close to a stationary point, so the tangent meets the x -axis far from the required root, and the sequence converges to the wrong root	B1(AO1.1b) B1(AO1.1b) [2] B1(AO2.3) B1(AO2.4) [2]	Correct first iteration x_2 and x_3 correct to at least 3dp Explanations do not need to include a sketch; if a sketch is included, ignore any inaccuracies if correct explanation is given; sketch with no explanation scores 0 'close to stationary point' oe seen 'converges to wrong root' oe seen	
		d	Choose starting value near the root and not near a stationary point, eg take $x_0 = 2$	E1(AO2.1) [1]		
			Total	9		

Question			Answer/Indicative content	Marks	Part marks and guidance	
11			When $x = 0$ $e^0 - 5 \times 0^3 = 1 > 0$ When $x = 1$ $e^1 - 5 \times 1^3 = e - 5 < 0$ So [as the function is continuous and there is a change of sign] there is a root between 0 and 1	M1 (AO 1.1a) E1 (AO 2.2a) [2]	Attempting to evaluate the function at both values Conclusion from correct values **<u>Examiner's Comments</u>** This question was generally well answered with only a few candidates making an arithmetical error that cost a mark.	
			Total	2		

Question			Answer/Indicative content	Marks	Part marks and guidance	
12		a	$f'(x) = \frac{1}{3}\left(27-8x^3\right)^{-\frac{2}{3}} \times (-24x^2)$ $\left[= \frac{-8x^2}{\left(27-8x^3\right)^{\frac{2}{3}}}\right]$ $f'(1.5) = -\frac{8\times1.5^2}{0}$ and dividing by zero gives the error.	M1 (AO1.1a) A1(AO 1.1) E1 (AO2.4) [3]	Using the chain rule Allow unsimplified Sufficient to say "can't divide by zero" oe **Examiner's Comments** This was well answered as most candidates used the chain rule successfully and realised that substituting x = 1.5 gives a zero in the denominator	

Question			Answer/Indicative content	Marks	Part marks and guidance
		b	$\left(27-8x^3\right)^{1/3}=27^{1/3}\left(1-\frac{8}{27}x^3\right)^{1/3}$ $=3\left(1+\left(\frac{1}{3}\right)\left(-\frac{8x^3}{27}\right)+\frac{\left(\frac{1}{3}\right)\left(-\frac{2}{3}\right)}{2!}\left(-\frac{8x^3}{27}\right)^2+\ldots\right)$ $=3-\frac{8x^3}{27}-\frac{64x^6}{2187}+\ldots$	B1 (AO 3.1a) M1 (AO 1.1a) A1 (AO 1.1b) [1]	Dealing with the 27 correctly Using the Binomial expansion substantially correctly Cao **<u>Examiner's Comments</u>** Many candidates dealt successfully with the 27, but when that was done without clear working shown, could cost 2 marks here. Some candidates simplified their coefficients early and incorrectly, so it was not always clear that the binomial expansion had been used, costing the method mark. **AfL** Make your method clear by writing down the factors of each term before simplifying.

Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	The binomial expansion is valid for $\left\|-8\frac{x^3}{27}\right\|<1$ $\|x\|$ <1.5 and the limits of the integral are completely in this interval.	B1 (AO 2.4) E1 (AO 2.3) [2]	Allow unsimplified but must use correct modulus notation or equivalent Must indicate that the limits of the integral lie in their interval for which the expansion is valid. **<u>Examiner's Comments</u>** One of the assessment objectives in the specification is to test the ability of a candidate to assess the validity of an argument as in this question. Not many candidates realised that the key to this explanation was to find the range of values for which the binomial expansion is valid. The limits lie well within the valid range so the method is valid.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		d	$\frac{0.25}{2}(3+2.6684+2(2.9954+2.9625+2.8694))$ $=\frac{0.25}{2}\times 23.3224=2.9153$	B1 (AO 1.1a) M1 (AO 1.1b) A1 (AO 1.1b) [3]	*h* = 0.25 used For sum in the bracket – condone one slip. Allow for 2.92 or better **<u>Examiner's Comments</u>** This was generally done very well.	Values from candidates own calculators may differ in the last decimal place.
		e	There is area between the curve and the top of the trapezia, so some area is missing from the estimate.	E1 (AO 2.4) [1]	Allow for any sensible explanation eg the trapezia are under the curve **<u>Examiner's Comments</u>** Most candidates were able to explain this clearly. Some had learned that a curve being concave downwards would give an underestimate but gave no indication as to why that would be, so lost the mark. AfL Clear annotated sketches can support a mathematical explanation better than an extended written response.	"The curve is concave downwards" on its own is not quite enough
			Total	12		

Question			Answer/Indicative content	Marks	Part marks and guidance	
13		a	$1 - 0.5^2$	B1 (AO 2.2a) [1]	oe	
		b	$\frac{0.5}{2}\{(2.718282+0.049787)+2(2.117+1+0.286505)\}$ = 2.39 (correct to 3 significant figures)	M1 (AO 1.1a) A1 (AO 1.1) [2]		
		c	$\frac{dy}{dx} = -2xe^{1-x^2}$ $\frac{dy}{dx} = 0$ only when $x = 0$, giving $y = e^{1-0} = e$	M1 (AO 1.1a) E1 (AO 2.1) [2]	AG Convincing completion	
			Total	5		

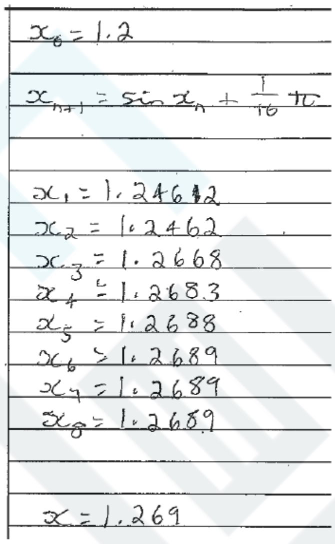
Question			Answer/Indicative content	Marks	Part marks and guidance	
14		a	Using $\sin\theta \approx \theta$ with $\theta = 2x^2$ and $\cos\theta \approx 1-\frac{1}{2}\theta^2$ with $\theta = 5x$ gives $2x^2 - \left(1-\frac{1}{2}(5x)^2\right) = 0$ $\frac{29}{2}x^2 = 1 \Rightarrow x = \sqrt{\frac{2}{29}} = 0.2626\ldots$ So estimate is 0.26 to 2 decimal places	M1 (AO 2.1) M1 (AO 2.1) A1 (AO 2.1) [3]	Allow slip in $(5x)^2$ Attempt to solve for x AG Must be rounded to 2 dp following either exact answer or answer to more than 2 dp seen	
		b	Denoting $\sin 2x^2 - \cos 5x$ by $f(x)$: $f(0.255) = -0.1618\ldots$ and $f(0.265) = -0.1033\ldots$ These are both negative so the root does not lie between 0.255 and 0.265, so the estimate is not correct to 2 decimal places	M1 (AO 2.1) A1 (AO 2.2a) [2]	Search for sign change using their 2dp value ±0.005 Complete argument from correct figs	oe, e.g. comparing values of $\sin 2x^2$ and $\cos 5x$ at end-points

Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	$f'(x) = 4x\cos 2x^2 + 5\sin 5x$ $x_1 = 0.2 - \frac{\sin(2\times0.2^2)-\cos(5\times0.2)}{4\times0.2\cos(2\times0.2^2)+5\sin(5\times0.2)}$ $= 0.2 - \frac{-0.460\,387\,6119}{5.004\,796\,289} = 0.291\,989\,28\ldots$ so x_1 = 0.291 989 3 correct to 7 dp	M1 (AO 1.1a) M1 (AO 1.1a) A1 (AO 1.1) M1 (AO 2.1) A1 (AO 2.1) [5]	Attempt to find $f'(x)$ Use of chain rule for $\sin 2x^2$ Derivative fully correct Use of iterative formula to find x_1 AG Answer must follow from clear working	
		d	x_2 = 0.282 338 3 x_3 = 0.282 285 3 Root is 0.282 correct to 3sf	M1 (AO 1.1a) A1 (AO 1.1) B1 (AO 2.2b) [3]	Finds at least one more iteration For correct x_3	
			Total	13		

Question			Answer/Indicative content	Marks	Part marks and guidance		
15		a	$h = 0.5 \Rightarrow$ Integral $\approx$ $\frac{1}{2} \times 0.5 \times 16.2075$ $= 4.051\ 875$	M1 (AO 1.1a) A1 (AO 1.1b) [2]	Substitution of h and the total from spreadsheet and using it in the trapezium rule formula awrt 4.05	Allow recalculation of the spreadsheet total from scratch	
		b	The estimate is an overestimate; as the curve is concave upwards the tops of the trapezia are above the curve and so the trapezia include extra area	E1 (AO 2.2a) [1]	Overestimate stated with clear explanation (must include reference to trapezia being above the curve, or a suitable diagram showing this)	Do not allow for argument based on a value for the integral found by calculator	
			Total	3			

Question			Answer/Indicative content	Marks	Part marks and guidance
16	a		Sector area $=\frac{1}{2}r^2x$ and Triangle $=\frac{1}{2}r^2$ sin x Area segment $=\frac{1}{2}r^2$ (x –sin$x$) $\frac{1}{2}r^2(x - \sin x) = 0.05\pi r^2$ $x - \sin x = 2 \times 0.05 \times \pi \Rightarrow x$ $- \sin x - \frac{1}{10}\pi = 0$	M1 (AO 2.1) A1 (AO 2.1) B1 (AO 2.1) A1 (AO 2.1) [4]	Both areas seen segment area found $0.05\pi r^2$ oe seen **AG** Must be fully shown and correct rear rangement **Examiner's Comments** For candidates who clearly labelled the areas of the regions, this was quite a straightforward proof. Errors usually came from omitting the square for the radius or using incorrect formulae. Some candidates looked to the given answer and tried to get the 5% to give the fraction 1/10.

Question			Answer/Indicative content	Marks	Part marks and guidance	
	b		$x_{n+1} = x_n - \dfrac{x_n - \sin x_n - \frac{1}{10}\pi}{1 - \cos x_n}$	B1 (AO 1.1a) [1]	x_{n+1} = must be seen Algebraic form must be seen Derivative must be worked out **Examiner's Comments** The generic formula for Newton-Raphson iteration is given in the formula pages, so candidates should not have expected that all that was required was copying this and doing nothing more. Some lost the mark for the formula by giving only the expression from the right-hand side, missing the x_{n+1} = to complete the formula. Where the formula only appeared in part (c), the mark for part (b) was not given retrospectively unless the candidate indicated where their answer was to be found.	Condone x used instead of x_n in the fraction part.
	c		$x_0 = 1.2$ $x = 1.27245\ldots$ $x = 1.26895\ldots$ $x = 1.26894\ldots$ So root is 1.269 to 3 dp	M1 (AO 1.1a) A1 (AO1.1b) A1 (AO 2.2b) [3]	3 iterations recorded The first 3 iterations correct Allow 1.27, 1.269 or more decimal places if correct. **Examiner's Comments** A calculator set in degrees	Root is 1.2 68947865 to 9 dp

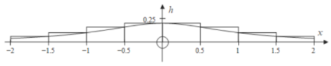
Question			Answer/Indicative content	Marks	Part marks and guidance	
					gave a sequence that did not converge, but many candidates did not realise how to fix the issue. Some candidates who had a converging sequence did not use their answers to give a value for the root. **Exemplar 4**  $x_0 = 1.2$ $x_{n+1} = \sin x_n + \frac{1}{10}\pi$ $x_1 = 1.24642$ $x_2 = 1.2462$ $x_3 = 1.2668$ $x_4 \doteq 1.2683$ $x_5 = 1.2688$ $x_6 = 1.2689$ $x_7 = 1.2689$ $x_8 = 1.2689$ $x = 1.269$ The question asks for the Newton-Raphson method to be used, yet in this example the $x = g(x)$ method is used. The method mark in part (c) was awarded for three terms of whichever sequence the candidate used but the first accuracy mark was not awarded since these are not the correct terms from the Newton-Raphson method. The second accuracy mark was given here as the sequence is seen converging to the correct root.	
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
17			any rearrangement to obtain $x = g(x)$ from given derivative = 0 $x = \frac{3 - \ln x}{2}$ use of their $g(x_n) = \frac{3 - \ln x_n}{2}$ to obtain at least two iterates eg 2, 1.1534, 1.4286... 1.3500 cao ------------------------------ *Alternatively,* $x_{n+1} = x_n - \frac{2x_n - 3 + \ln x_n}{\text{their } \left(2 + \frac{1}{x_n}\right)}$ use of their N-R formula to obtain x_1, x_2,...eg 1.5, 1.34795, 1.34996,...1.349962 1.3500 cao	M1* (AO2.1) A1 (AO1.1) M1dep* (AO1.1) A1 (AO2.2a) [4] M1* (AO2.1) A1 (AO1.1) M1dep* (AO1.1) A1 (AO2.2a) [4]	allow sign error any correct rearrangement must see iterates Newton-Raphson iterative formula seen (not for solving $f(x) = 0$) formula all correct must see iterates **Examiner's Comments** Candidates who did well in this question used either the Newton-Raphson method or fixed point iteration. They showed the iterates and used their calculators efficiently to achieve the specified 4	eg $x = e^{2x-3}$ need not see subscripts in iterative formula **0** for 1.3500 unsupported trial and improvement does not score need not see subscripts in iterative formula

Question			Answer/Indicative content	Marks	Part marks and guidance	
					decimal place accuracy. Candidates who did less well made slips in their iterative formulae or (quite commonly) gave the final answer as 1.3450. Candidates who struggled with this question solved $f(x) = 0$ instead of $\frac{dy}{dx} = 0$ or used the equation solver or a non-iterative method to find the root.	
			Total	4		

Question	Answer/Indicative content	Marks	Part marks and guidance	
18	$\int\left(\frac{1}{(x+2)}-\frac{1}{(x+3)}\right)\mathrm{d}x$ = ln $\lvert x+2\rvert$ – ln $\lvert x+3\rvert$ [+c] $\ln\left\lvert\frac{x+2}{x+3}\right\rvert+c$	M2 (AO1.1a) (AO1.1) A1 (AO2.2a) [3]	For both terms correct FT their fractions **M1** for one of their terms correct **A0** if modulus sign missing or if further work eg to try and find c	Need brackets or better (eg single fraction Ignore ‘ $f(x)$ = ’
			<u>Examiner's Comments</u> In (b) most could integrate the reciprocals correctly to get natural logarithms but only the strongest realised that modulus should be used as negatives are not part of the domain for log functions Exemplar 3 $\int\frac{1}{x+2}\,dx-\int\frac{1}{x+3}\,dx.=\int\frac{1}{(x+2)(x+3)}\,dx.$ $=\ln(x+2)-\ln(x+3)+c.$ $\quad \ln a-\ln b=\ln\frac{a}{b}.$ $=\ln\left(\frac{x+2}{x+3}\right)+c$ This was probably the most common response we saw to this part. The candidate has correctly integrated each fraction to get ln and also uses a log law correctly. If they had used modulus bars instead of round brackets they would have earned the last mark as well,	

Question			Answer/Indicative content	Marks	Part marks and guidance	
			Total	3		
19	a		**DR** $\ln 3^2$, $\ln 3^3$ seen $\ln 3 \times 2\ln 3 \times 3\ln 3 = 6(\ln 3)^3$	B1 (AO1.1a) B1 (AO2.2a) [2]	<u>Examiner's Comments</u> With this being a detailed reasoning question, it was expected that candidates show enough steps in their answers to provide detail of why each step follows. In this part, many went straight to $\ln 3 \times 2\ln 3 \times 3\ln 3$ without showing the intermediate step. This meant that those who got to $6(\ln 3)^3$ only scored 1 mark. Other fairly common mistakes were in misplacing the brackets in the final answer.	
	b		**DR** $3 > e$ so $\ln 3 > 1$ $(\ln 3)^3 > 1$ so $6(\ln 3)^3 > 6$	M1 (AO2.2a) E1 (AO2.4) [2]	Must mention e Convincing completion (answer given) <u>Examiner's Comments</u> Not very many candidates were successful in this part, with only a small proportion even using the fact that e < 3. Many tried finding numerical answers via their calculators rather than working with e.	
			Total	4		

<table>
<tr><th colspan="3">Question</th><th>Answer/Indicative content</th><th>Marks</th><th colspan="2">Part marks and guidance</th></tr>
<tr><td>20</td><td>a</td><td></td><td>
<table><tr><td>x</td><td>0</td><td>±0.5</td><td>±1</td><td>±1.5</td><td>2</td></tr><tr><td>h</td><td>$\frac{1}{5}$</td><td>$\frac{4}{25}$</td><td>$\frac{1}{10}$</td><td>$\frac{4}{65}$</td><td>$\frac{1}{25}$</td></tr><tr><td>h</td><td>0.2</td><td>0.16</td><td>0.1</td><td>0.061538</td><td>0.04</td></tr></table>
= 2 × 0.5(0.2 + 0.16 + 0.1 + 0.061538)

=0.522 to 3sf</td><td>M1(AO3.4)

M1(AO1.1a)
A1(AO1.1)
[3]</td><td>Finding at least 2 distinct values for h Need not be a table of values

Using rectangles forming UB for area with width 0.5. Need not be drawn</td><td>Allow both M marks if only half the area considered.</td></tr>
<tr><td></td><td>b</td><td></td><td>Area $\approx \frac{0.5}{2}\left(0.2+0.04+2(0.16+0.1+0.061538)\right)$
=0.221 (to 3 sf)</td><td>M1(AO1.1a)
A1(1.1)
[2]</td><td>Using trapezium rule with 5 h values FT incorrect h values for the M mark Allow awrt 0.22</td><td>Using the definite integral functionality of calculator gives 0.2214297 … So method must be seen.</td></tr>
<tr><td></td><td>c</td><td></td><td>Volume = 2 × (0.2208 × 10)

= 4.42 m^3</td><td>M1(AO1.1a)

A1(AO1.1)
[2]</td><td>Condone missing factor of 2 for method mark
FT part (b)</td><td></td></tr>
<tr><td></td><td></td><td></td><td>Total</td><td>7</td><td colspan="2"></td></tr>
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